



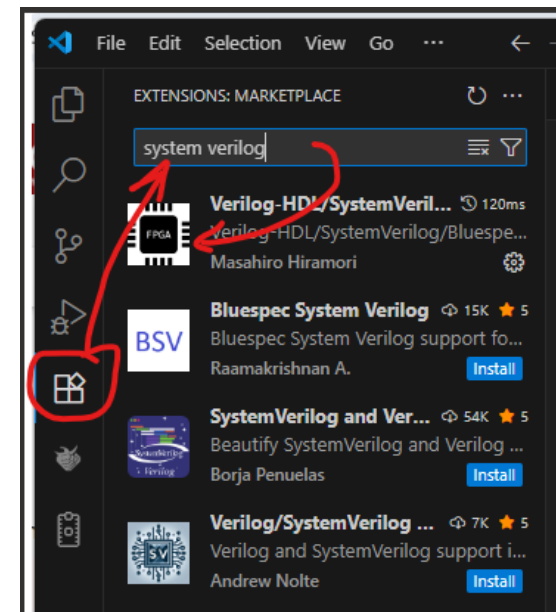
Exercise 04

Generate Verilog DUT using VS Code & Copilot

- We provide **counter_tb.sv** – a testbench for a 4-bit up/down counter with enable.
 - It instantiates the DUT, applies stimulus, and self-checks the results.
 - Look over the code to make sure you understand what is going on.
- Take a few moment to think about what the DUT code should look like, roughly.
- We will experiment with prompting Copilot to generate the DUT code.

- **VS Code & Copilot set up**

- Make sure you sign up for a Github education account.
 - Launch VS Code, sign into your Github account, and install plugin for system verilog support (select the one by Masahiro Hiramori)
- Create a new file called **counter.sv**
 - Having the extension .sv should make it recognize this as a system Verilog file.





Exercise 04

Generate Verilog DUT using VS Code & CoPilot

- Using comments `//` at the beginning of the file, prompt copilot to generate the DUT code.
- Some things to specify:
 - 4-bit up/down counter with enable
 - With signals **clk**, **rst_n**, **en**, and **up_dwn_n**
 - Output called **cnt** should count up when **up_dwn_n** is high and down when it is low.
- The resulting code should be an always block with some nested if/else clauses inside it.
- Did copilot give you good quality code right off the bat? Or did you have to tweak your prompt and iterate a few times in order to get what you want?

→ Submit counter.sv